

# The Electric Vehicle Routing Problem with Overnight Depot Charging and En-Route Charging

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## Appendix A. Summary of Symbols

Table A.1: Sets, parameters, and decision variables of the proposed MILP

| Set                                      | Description                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| $C$                                      | set of customer nodes                                                                                     |
| $F$                                      | set of charging station nodes                                                                             |
| $N$                                      | set of all nodes (depot, customers and charging stations)                                                 |
| $A$                                      | set of arcs                                                                                               |
| $\mathcal{P}$                            | set of paths between any two nodes                                                                        |
| $\mathcal{P}^{\text{DC}}$                | set of depot-to-customer paths                                                                            |
| $\mathcal{P}^{\text{CC}}$                | set of customer-to-customer paths                                                                         |
| $\mathcal{P}^{\text{CD}}$                | set of customer-to-depot paths                                                                            |
| $\mathcal{P}^{\text{D}}$                 | set of direct paths                                                                                       |
| $\mathcal{P}^{\text{CH}}$                | set of charging paths                                                                                     |
| $\mathcal{L}_p$                          | index set $\{1, \dots, n(p)\}$ of public-CS visits along path $p \in \mathcal{P}^{\text{CH}}$             |
| $O$                                      | set of all overnight charging operations on the path set $\mathcal{P}^{\text{DC}}$                        |
| $\mathcal{A}^{\text{OC}}$                | retained sequencing arcs between the dummy source/sink and real depot charging operations                 |
| Parameter                                | Description                                                                                               |
| $q_i$                                    | demand of customer $i$                                                                                    |
| $s_i$                                    | service time of customer $i$                                                                              |
| $[\alpha_i, \beta_i]$                    | time window of customer $i$                                                                               |
| $T$                                      | end of the planning horizon                                                                               |
| $T_B$                                    | earliest allowable departure time for daytime routing                                                     |
| $[E_{\min}, E_{\max}]$                   | prescribed operating SOC range for EV                                                                     |
| $Q$                                      | load capacity of EV                                                                                       |
| $\rho_f$                                 | energy charging rate at public CSs (energy per unit of model time)                                        |
| $\rho_s$                                 | energy charging rate at depot chargers (energy per unit of model time)                                    |
| $\mathcal{B}$                            | number of available depot chargers                                                                        |
| $d_{ij}$                                 | travel distance from node $i$ to node $j$ , used by the large-scale ReplacePath rule                      |
| $t_{ij}, c_{ij}$                         | travel time and energy consumption between $(i, j)$                                                       |
| $\delta_p^-, \delta_p^+$                 | origin and destination of path $p \in \mathcal{P}$                                                        |
| $n(p)$                                   | number of CSs visited along path $p \in \mathcal{P}$                                                      |
| $e_p$                                    | energy consumption incurred when traversing the path $p \in \mathcal{P}$                                  |
| $t_p$                                    | travel time (excluding any charging time) when traversing the path $p \in \mathcal{P}$                    |
| $cs(p, \ell)$                            | $\ell^{\text{th}}$ CS visited along path $p \in \mathcal{P}^{\text{CH}}$                                  |
| $(o^-, o^+)$                             | source and sink dummy operations of the depot-charging sequencing network                                 |
| Variable                                 | Description                                                                                               |
| $x_p$                                    | 1 if path $p \in \mathcal{P}$ is selected; 0 otherwise                                                    |
| $\tau_p$                                 | departure time from the origin node of path $p$                                                           |
| $r_p$                                    | departure SOC from the origin node of path $p$                                                            |
| $\underline{r}_{p\ell}, \bar{r}_{p\ell}$ | arriving SOC and departing SOC of the EV at $\ell^{\text{th}}$ CS on path $p \in \mathcal{P}^{\text{CH}}$ |
| $\Delta_p$                               | total charging time incurred along path $p \in \mathcal{P}^{\text{CH}}$                                   |
| $\underline{t}_p, \bar{t}_p$             | start time and end time of the overnight charging operation for path $p \in \mathcal{P}^{\text{DC}}$      |
| $f_{o'o}$                                | 1 if operation $o'$ immediately precedes operation $o$ on the same charger; 0 otherwise                   |
| $u_i$                                    | remaining vehicle load after serving customer $i$                                                         |

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## Appendix B. Detailed results for the EVRP-OC instances

Table B.1: Detailed results on small-scale instances under  $\mathcal{B} = 1$

| Instance | MILP    |        |          | Matheuristic |         |          |     |        |     |
|----------|---------|--------|----------|--------------|---------|----------|-----|--------|-----|
|          | Obj     | Gap    | Time (s) | Best 10      | Avg 10  | Time (s) | #CS | Dist.  | #NV |
| C101C5   | 907.19  | 0.00%  | 1.70     | 907.19       | 908.30  | 0.13     | 3   | 264.44 | 2   |
| C103C5   | 394.96  | 0.00%  | 0.43     | 394.96       | 394.96  | 0.17     | 1   | 172.97 | 2   |
| C206C5   | 572.76  | 0.00%  | 10.13    | 572.76       | 572.76  | 0.21     | 2   | 244.37 | 2   |
| C208C5   | 425.39  | 0.00%  | 1.52     | 425.39       | 425.39  | 0.16     | 2   | 205.67 | 2   |
| R104C5   | 182.93  | 0.00%  | 0.99     | 182.93       | 182.93  | 0.16     | 2   | 149.26 | 2   |
| R105C5   | 226.51  | 0.00%  | 0.79     | 226.51       | 226.51  | 0.20     | 3   | 177.49 | 2   |
| R202C5   | 162.17  | 0.00%  | 1.99     | 162.17       | 162.17  | 0.23     | 2   | 128.78 | 1   |
| R203C5   | 232.07  | 0.00%  | 1.17     | 232.07       | 232.07  | 0.30     | 3   | 195.63 | 2   |
| RC105C5  | 294.28  | 0.00%  | 2.41     | 294.28       | 294.28  | 0.27     | 3   | 243.09 | 3   |
| RC108C5  | 335.98  | 0.00%  | 1.43     | 335.98       | 335.98  | 0.15     | 3   | 269.74 | 2   |
| RC204C5  | 193.06  | 0.00%  | 1.09     | 193.06       | 193.06  | 0.20     | 1   | 187.69 | 3   |
| RC208C5  | 203.17  | 0.00%  | 2.37     | 203.17       | 203.17  | 0.18     | 2   | 167.98 | 1   |
| C101C10  | INF     | —      | 10.77    | —            | —       | —        | —   | —      | —   |
| C104C10  | 948.41  | 30.00% | 7200.13  | 948.41       | 948.41  | 0.49     | 4   | 282.67 | 2   |
| C202C10  | 672.89  | 0.00%  | 3543.25  | 672.89       | 672.89  | 0.89     | 2   | 275.39 | 3   |
| C205C10  | 955.62  | 0.00%  | 550.01   | 955.62       | 1083.25 | 0.60     | 4   | 285.98 | 2   |
| R102C10  | INF     | —      | 9.70     | —            | —       | —        | —   | —      | —   |
| R103C10  | 267.84  | 0.00%  | 183.58   | 267.84       | 267.84  | 0.85     | 2   | 210.79 | 3   |
| R201C10  | 261.51  | 0.00%  | 12.57    | 261.51       | 261.51  | 1.29     | 2   | 228.36 | 3   |
| R203C10  | 287.28  | 0.00%  | 26.54    | 287.28       | 287.28  | 0.78     | 3   | 232.68 | 2   |
| RC102C10 | INF     | —      | 0.10     | —            | —       | —        | —   | —      | —   |
| RC108C10 | INF     | —      | 1173.91  | —            | —       | —        | —   | —      | —   |
| RC201C10 | 381.19  | 0.00%  | 13.96    | 381.19       | 381.19  | 1.45     | 4   | 312.57 | 3   |
| RC205C10 | 392.46  | 0.00%  | 2.67     | 392.46       | 392.46  | 1.32     | 3   | 325.98 | 2   |
| C103C15  | NFS     | —      | 7200.09  | 1668.12      | 1668.12 | 1.04     | 8   | 433.54 | 3   |
| C106C15  | NFS     | —      | 7200.09  | NFS          | NFS     | 0.71     | NFS | NFS    | NFS |
| C202C15  | NFS     | —      | 7200.06  | NFS          | NFS     | 1.52     | NFS | NFS    | NFS |
| C208C15  | 1100.56 | 41.19% | 7200.08  | 1100.56      | 1100.56 | 1.82     | 4   | 341.54 | 3   |
| R102C15  | INF     | —      | 1892.36  | —            | —       | —        | —   | —      | —   |
| R105C15  | NFS     | —      | 7200.33  | NFS          | NFS     | 1.83     | NFS | NFS    | NFS |
| R202C15  | 485.22  | 0.00%  | 4122.05  | 485.22       | 485.22  | 4.86     | 6   | 378.14 | 4   |
| R209C15  | 374.53  | 0.00%  | 188.63   | 374.53       | 374.53  | 2.98     | 3   | 307.68 | 3   |
| RC103C15 | NFS     | —      | 7200.05  | NFS          | NFS     | 1.31     | NFS | NFS    | NFS |
| RC108C15 | NFS     | —      | 7200.06  | NFS          | NFS     | 1.13     | NFS | NFS    | NFS |
| RC202C15 | 472.92  | 0.00%  | 95.90    | 472.92       | 473.47  | 4.31     | 5   | 405.28 | 4   |
| RC204C15 | 380.05  | 8.45%  | 7200.06  | 380.05       | 380.21  | 1.37     | 4   | 312.00 | 2   |

INF: infeasibility proved by the MILP; NFS: no feasible solution found within the time limit; —: not available/not applicable. Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.2: Detailed results on small-scale instances under  $\mathcal{B} = 2$ 

| Instance | MILP    |        |          | Matheuristic |         |          |     |        |     |
|----------|---------|--------|----------|--------------|---------|----------|-----|--------|-----|
|          | Obj     | Gap    | Time (s) | Best 10      | Avg 10  | Time (s) | #CS | Dist.  | #NV |
| C101C5   | 491.57  | 0.00%  | 0.97     | 491.57       | 491.57  | 0.38     | 2   | 253.21 | 4   |
| C103C5   | 165.67  | 0.00%  | 0.95     | 165.67       | 165.67  | 0.25     | 0   | 165.67 | 3   |
| C206C5   | 331.28  | 0.00%  | 0.69     | 331.28       | 331.28  | 0.48     | 1   | 268.42 | 4   |
| C208C5   | 376.77  | 0.00%  | 1.51     | 376.77       | 376.77  | 0.24     | 2   | 205.00 | 2   |
| R104C5   | 146.17  | 0.00%  | 0.12     | 146.17       | 146.17  | 0.18     | 1   | 136.69 | 2   |
| R105C5   | 177.54  | 0.00%  | 0.11     | 177.54       | 177.54  | 0.35     | 2   | 156.08 | 2   |
| R202C5   | 162.17  | 0.00%  | 1.78     | 162.17       | 162.17  | 0.24     | 2   | 128.78 | 1   |
| R203C5   | 232.07  | 0.00%  | 1.36     | 232.07       | 232.07  | 0.32     | 3   | 195.63 | 2   |
| RC105C5  | 259.57  | 0.00%  | 1.08     | 259.57       | 261.03  | 0.45     | 3   | 238.18 | 3   |
| RC108C5  | 292.32  | 0.00%  | 0.63     | 292.32       | 292.32  | 0.29     | 3   | 253.93 | 2   |
| RC204C5  | 193.06  | 0.00%  | 1.41     | 193.06       | 193.06  | 0.25     | 1   | 187.69 | 3   |
| RC208C5  | 203.17  | 0.00%  | 1.18     | 203.17       | 203.17  | 0.23     | 2   | 167.98 | 1   |
| C101C10  | 1198.22 | 0.00%  | 47.14    | 1198.22      | 1207.97 | 1.27     | 5   | 405.15 | 4   |
| C104C10  | 684.89  | 4.36%  | 7200.11  | 684.89       | 684.89  | 1.05     | 3   | 273.93 | 2   |
| C202C10  | 317.01  | 0.00%  | 14.99    | 317.01       | 317.01  | 1.91     | 1   | 291.19 | 4   |
| C205C10  | 480.83  | 0.00%  | 22.21    | 480.83       | 480.83  | 1.10     | 2   | 228.28 | 2   |
| R102C10  | 304.93  | 0.00%  | 5.67     | 304.93       | 304.93  | 1.79     | 3   | 253.35 | 3   |
| R103C10  | 226.87  | 0.00%  | 21.10    | 226.87       | 226.87  | 1.48     | 1   | 207.87 | 4   |
| R201C10  | 252.10  | 0.00%  | 5.61     | 252.10       | 255.41  | 2.10     | 2   | 224.54 | 3   |
| R203C10  | 287.28  | 0.00%  | 21.67    | 287.28       | 287.28  | 1.07     | 3   | 232.68 | 2   |
| RC102C10 | INF     | –      | 13.50    | –            | –       | –        | –   | –      | –   |
| RC108C10 | 451.78  | 0.00%  | 30.00    | 451.78       | 451.78  | 2.16     | 4   | 387.65 | 4   |
| RC201C10 | 347.37  | 0.00%  | 3.57     | 347.37       | 349.92  | 2.83     | 3   | 310.06 | 3   |
| RC205C10 | 383.55  | 0.00%  | 2.24     | 383.55       | 383.55  | 1.94     | 3   | 341.38 | 3   |
| C103C15  | 1152.58 | 62.62% | 7200.17  | 1037.78      | 1037.78 | 3.87     | 5   | 368.60 | 3   |
| C106C15  | 701.43  | 19.48% | 7200.08  | 701.43       | 701.43  | 3.97     | 3   | 281.82 | 3   |
| C202C15  | 1029.65 | 42.49% | 7200.07  | 1029.65      | 1029.65 | 5.49     | 5   | 431.17 | 4   |
| C208C15  | 673.78  | 0.00%  | 372.63   | 673.78       | 673.78  | 4.08     | 3   | 351.26 | 4   |
| R102C15  | 586.13  | 0.00%  | 5341.15  | 586.13       | 595.76  | 5.89     | 8   | 441.46 | 5   |
| R105C15  | 468.44  | 0.00%  | 910.30   | 468.44       | 468.44  | 5.68     | 5   | 373.17 | 5   |
| R202C15  | 455.51  | 0.00%  | 823.25   | 455.51       | 455.51  | 6.39     | 6   | 378.14 | 4   |
| R209C15  | 374.53  | 0.00%  | 71.27    | 374.53       | 374.53  | 5.55     | 3   | 307.68 | 3   |
| RC103C15 | 473.22  | 0.00%  | 1191.10  | 473.22       | 473.22  | 5.64     | 4   | 397.75 | 5   |
| RC108C15 | 461.49  | 0.00%  | 3288.46  | 461.49       | 461.67  | 4.80     | 4   | 379.51 | 3   |
| RC202C15 | 462.88  | 0.00%  | 28.54    | 462.88       | 462.88  | 6.75     | 4   | 418.67 | 4   |
| RC204C15 | 371.05  | 4.47%  | 7200.21  | 371.05       | 371.05  | 1.69     | 4   | 310.58 | 2   |

INF: infeasibility proved by the MILP; NFS: no feasible solution found within the time limit; –: not available/not applicable. Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.3: Detailed results on small-scale instances under  $\mathcal{B} = 3$ 

| Instance | MILP   |        |          | Matheuristic |        |          |     |        |     |
|----------|--------|--------|----------|--------------|--------|----------|-----|--------|-----|
|          | Obj    | Gap    | Time (s) | Best 10      | Avg 10 | Time (s) | #CS | Dist.  | #NV |
| C101C5   | 348.97 | 0.00%  | 0.67     | 348.97       | 348.97 | 0.53     | 1   | 250.04 | 4   |
| C103C5   | 165.67 | 0.00%  | 0.84     | 165.67       | 165.67 | 0.25     | 0   | 165.67 | 3   |
| C206C5   | 331.28 | 0.00%  | 1.25     | 331.28       | 331.28 | 0.57     | 1   | 268.42 | 4   |
| C208C5   | 376.77 | 0.00%  | 3.21     | 376.77       | 376.77 | 0.25     | 2   | 205.00 | 2   |
| R104C5   | 146.17 | 0.00%  | 0.17     | 146.17       | 146.17 | 0.24     | 1   | 136.69 | 2   |
| R105C5   | 177.54 | 0.00%  | 0.09     | 177.54       | 177.54 | 0.42     | 2   | 156.08 | 2   |
| R202C5   | 162.17 | 0.00%  | 1.74     | 162.17       | 162.17 | 0.23     | 2   | 128.78 | 1   |
| R203C5   | 232.07 | 0.00%  | 1.32     | 232.07       | 232.07 | 0.35     | 3   | 195.63 | 2   |
| RC105C5  | 246.93 | 0.00%  | 0.36     | 246.93       | 246.93 | 0.44     | 1   | 238.05 | 3   |
| RC108C5  | 292.32 | 0.00%  | 0.54     | 292.32       | 292.32 | 0.30     | 3   | 253.93 | 2   |
| RC204C5  | 193.06 | 0.00%  | 0.94     | 193.06       | 193.06 | 0.28     | 1   | 187.69 | 3   |
| RC208C5  | 203.17 | 0.00%  | 1.58     | 203.17       | 203.17 | 0.26     | 2   | 167.98 | 1   |
| C101C10  | 888.52 | 0.00%  | 158.24   | 888.52       | 890.23 | 2.18     | 4   | 412.64 | 5   |
| C104C10  | 575.70 | 0.00%  | 323.25   | 575.70       | 575.70 | 1.67     | 3   | 309.86 | 3   |
| C202C10  | 317.01 | 0.00%  | 11.89    | 317.01       | 317.01 | 2.90     | 1   | 291.19 | 4   |
| C205C10  | 374.46 | 0.00%  | 4.86     | 374.46       | 374.46 | 1.75     | 2   | 260.52 | 3   |
| R102C10  | 289.56 | 0.00%  | 1.81     | 289.56       | 289.56 | 2.25     | 2   | 262.92 | 4   |
| R103C10  | 211.14 | 0.00%  | 9.73     | 211.14       | 211.14 | 1.77     | 1   | 207.87 | 4   |
| R201C10  | 250.33 | 0.00%  | 4.91     | 250.33       | 250.33 | 2.65     | 2   | 224.54 | 3   |
| R203C10  | 287.28 | 0.00%  | 23.79    | 287.28       | 287.28 | 1.27     | 3   | 232.68 | 2   |
| RC102C10 | 498.14 | 0.00%  | 3.68     | 498.14       | 498.14 | 3.45     | 4   | 429.66 | 4   |
| RC108C10 | 393.23 | 0.00%  | 4.30     | 393.23       | 393.23 | 2.69     | 3   | 345.93 | 3   |
| RC201C10 | 340.01 | 0.00%  | 2.71     | 340.01       | 340.01 | 3.03     | 3   | 310.06 | 3   |
| RC205C10 | 383.55 | 0.00%  | 2.18     | 383.55       | 383.55 | 2.49     | 3   | 341.38 | 3   |
| C103C15  | 738.31 | 34.51% | 7200.06  | 738.31       | 738.31 | 6.32     | 5   | 369.54 | 4   |
| C106C15  | 440.31 | 0.00%  | 21.94    | 440.31       | 455.18 | 6.38     | 1   | 306.49 | 5   |
| C202C15  | 764.99 | 7.45%  | 7200.07  | 764.99       | 764.99 | 9.76     | 3   | 491.04 | 6   |
| C208C15  | 611.95 | 0.00%  | 127.07   | 611.95       | 617.90 | 6.59     | 3   | 372.14 | 4   |
| R102C15  | 510.09 | 0.00%  | 289.13   | 510.09       | 511.36 | 10.59    | 6   | 412.78 | 5   |
| R105C15  | 398.58 | 0.00%  | 37.96    | 398.58       | 399.07 | 9.15     | 4   | 336.94 | 4   |
| R202C15  | 444.59 | 0.00%  | 404.75   | 444.59       | 445.25 | 7.90     | 6   | 378.14 | 4   |
| R209C15  | 374.53 | 0.00%  | 90.60    | 374.53       | 374.53 | 6.32     | 3   | 307.68 | 3   |
| RC103C15 | 449.05 | 0.00%  | 230.99   | 449.05       | 450.82 | 7.87     | 4   | 397.75 | 5   |
| RC108C15 | 423.68 | 0.00%  | 362.90   | 423.68       | 423.68 | 5.83     | 4   | 370.25 | 3   |
| RC202C15 | 462.88 | 0.00%  | 27.84    | 462.88       | 463.11 | 8.18     | 4   | 418.67 | 4   |
| RC204C15 | 371.05 | 4.45%  | 7200.22  | 371.05       | 371.05 | 2.69     | 4   | 310.58 | 2   |

Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.4: Detailed Results for C1- and C2-Type Large-Scale Instances with Different Numbers of Depot Chargers

| Instance | $\mathcal{B}$ | Obj (Best 10) | Obj (Avg 10) | Time (min) | #NV | #CS | Dep. SOC | Travel Time | Charging Time |
|----------|---------------|---------------|--------------|------------|-----|-----|----------|-------------|---------------|
| C101     | 12            | 1589.68       | 1589.68      | 3.61       | 12  | 9   | 95.47%   | 1067.11     | 522.57        |
|          | 10            | 2001.19       | 2036.58      | 4.08       | 13  | 10  | 79.90%   | 1095.08     | 906.11        |
|          | 8             | 2580.26       | 2653.76      | 4.13       | 14  | 13  | 62.13%   | 1123.02     | 1457.24       |
| C102     | 12            | 1534.05       | 1534.05      | 3.62       | 12  | 8   | 92.34%   | 1031.33     | 502.73        |
|          | 10            | 1787.50       | 1850.94      | 4.04       | 14  | 8   | 78.03%   | 1079.45     | 708.05        |
|          | 8             | 2530.10       | 2612.76      | 4.29       | 13  | 13  | 65.52%   | 1100.47     | 1429.63       |
| C103     | 12            | 1526.79       | 1527.98      | 3.66       | 12  | 8   | 90.79%   | 1018.23     | 508.56        |
|          | 10            | 1667.43       | 1701.41      | 3.99       | 12  | 7   | 87.16%   | 1023.47     | 643.96        |
|          | 8             | 2324.86       | 2439.54      | 4.31       | 13  | 11  | 69.34%   | 1084.14     | 1240.72       |
| C104     | 12            | 1345.35       | 1346.52      | 3.41       | 11  | 7   | 89.60%   | 912.96      | 432.40        |
|          | 10            | 1350.65       | 1366.20      | 3.81       | 10  | 8   | 95.53%   | 895.53      | 455.12        |
|          | 8             | 2015.56       | 2213.91      | 4.97       | 15  | 8   | 67.62%   | 1083.33     | 932.23        |
| C105     | 12            | 1555.15       | 1555.15      | 3.67       | 12  | 8   | 93.58%   | 1045.27     | 509.88        |
|          | 10            | 1928.96       | 1960.63      | 4.12       | 13  | 8   | 79.06%   | 1061.46     | 867.50        |
|          | 8             | 2465.43       | 2532.75      | 4.36       | 14  | 12  | 63.58%   | 1109.33     | 1356.10       |
| C106     | 12            | 1553.70       | 1553.70      | 3.74       | 12  | 8   | 93.43%   | 1043.81     | 509.88        |
|          | 10            | 1859.02       | 1891.20      | 4.16       | 13  | 8   | 80.66%   | 1068.73     | 790.29        |
|          | 8             | 2460.63       | 2602.11      | 4.51       | 13  | 11  | 65.33%   | 1083.11     | 1377.52       |
| C107     | 12            | 1554.19       | 1554.19      | 3.60       | 12  | 8   | 92.99%   | 1040.74     | 513.44        |
|          | 10            | 1876.15       | 1919.08      | 3.82       | 13  | 8   | 79.62%   | 1064.32     | 811.84        |
|          | 8             | 2394.94       | 2432.50      | 4.43       | 13  | 11  | 65.30%   | 1067.95     | 1326.99       |
| C108     | 12            | 1547.99       | 1547.99      | 3.69       | 12  | 8   | 92.65%   | 1036.80     | 511.19        |
|          | 10            | 1781.74       | 1867.97      | 4.17       | 13  | 10  | 80.97%   | 1053.57     | 728.17        |
|          | 8             | 2478.88       | 2557.81      | 4.42       | 13  | 13  | 66.56%   | 1097.13     | 1381.75       |
| C109     | 12            | 1421.27       | 1425.21      | 3.80       | 11  | 7   | 92.65%   | 950.90      | 470.38        |
|          | 10            | 1780.66       | 1822.70      | 4.23       | 11  | 9   | 90.25%   | 1016.51     | 764.15        |
|          | 8             | 2305.99       | 2399.54      | 4.52       | 13  | 10  | 67.19%   | 1062.77     | 1243.22       |
| C201     | 4             | 958.18        | 966.21       | 2.84       | 7   | 3   | 83.60%   | 773.38      | 184.80        |
|          | 3             | 1271.03       | 1287.08      | 3.49       | 7   | 5   | 65.04%   | 761.91      | 509.11        |
|          | 2             | 1740.73       | 1768.99      | 3.94       | 8   | 8   | 42.48%   | 810.20      | 930.54        |
| C202     | 4             | 945.78        | 949.27       | 2.82       | 7   | 3   | 85.37%   | 779.79      | 165.99        |
|          | 3             | 1243.67       | 1273.84      | 3.53       | 7   | 4   | 68.05%   | 770.91      | 472.77        |
|          | 2             | 1580.20       | 1596.57      | 3.70       | 6   | 6   | 59.21%   | 773.91      | 806.28        |
| C203     | 4             | 929.92        | 936.94       | 2.84       | 7   | 2   | 86.13%   | 779.02      | 150.90        |
|          | 3             | 1147.34       | 1161.68      | 3.02       | 7   | 4   | 72.34%   | 765.95      | 381.39        |
|          | 2             | 1446.54       | 1454.50      | 3.46       | 6   | 6   | 61.42%   | 743.88      | 702.66        |
| C204     | 4             | 867.21        | 880.04       | 2.84       | 7   | 3   | 87.60%   | 767.76      | 99.45         |
|          | 3             | 1161.69       | 1180.68      | 3.14       | 7   | 3   | 83.18%   | 831.82      | 329.87        |
|          | 2             | 1431.70       | 1442.80      | 3.75       | 7   | 5   | 58.04%   | 769.17      | 662.53        |
| C205     | 4             | 952.05        | 956.61       | 2.96       | 6   | 3   | 86.20%   | 713.36      | 238.70        |
|          | 3             | 1232.27       | 1239.70      | 3.46       | 7   | 4   | 63.68%   | 739.97      | 492.30        |
|          | 2             | 1502.31       | 1516.92      | 3.77       | 6   | 6   | 54.27%   | 723.59      | 778.72        |
| C206     | 4             | 947.17        | 952.00       | 2.93       | 6   | 3   | 89.65%   | 728.54      | 218.62        |
|          | 3             | 1190.83       | 1207.16      | 3.28       | 6   | 4   | 75.42%   | 732.70      | 458.12        |
|          | 2             | 1483.33       | 1493.04      | 3.69       | 6   | 5   | 53.56%   | 714.12      | 769.21        |
| C207     | 4             | 923.35        | 944.03       | 2.88       | 7   | 3   | 87.12%   | 780.11      | 143.24        |
|          | 3             | 1205.78       | 1229.98      | 3.52       | 7   | 4   | 67.29%   | 752.25      | 453.52        |
|          | 2             | 1510.30       | 1515.83      | 3.79       | 7   | 5   | 48.05%   | 734.52      | 775.78        |
| C208     | 4             | 902.97        | 918.52       | 2.91       | 6   | 3   | 87.52%   | 704.51      | 198.46        |
|          | 3             | 1150.81       | 1175.15      | 3.16       | 7   | 3   | 71.11%   | 757.45      | 393.36        |
|          | 2             | 1449.61       | 1470.47      | 3.42       | 6   | 4   | 56.12%   | 716.40      | 733.21        |

Table B.5: Detailed Results for R1- and R2-Type Large-Scale Instances with Different Numbers of Depot Chargers

| Instance | $\mathcal{B}$ | Obj (Best 10) | Obj (Avg 10) | Time (min) | #NV | #CS | Dep. SOC | Travel Time | Charging Time |
|----------|---------------|---------------|--------------|------------|-----|-----|----------|-------------|---------------|
| R101     | 15            | 2062.80       | 2087.37      | 7.67       | 21  | 23  | 77.02%   | 1719.76     | 343.05        |
|          | 12            | NFS           | NFS          | 6.86       | –   | –   | –        | –           | –             |
|          | 9             | NFS           | NFS          | 6.45       | –   | –   | –        | –           | –             |
| R102     | 15            | 1776.25       | 1786.59      | 6.34       | 18  | 20  | 86.93%   | 1515.53     | 260.72        |
|          | 12            | 1877.73       | 1940.37      | 6.60       | 19  | 23  | 68.97%   | 1532.82     | 344.91        |
|          | 9             | NFS           | NFS          | 5.89       | –   | –   | –        | –           | –             |
| R103     | 15            | 1424.88       | 1435.87      | 5.51       | 15  | 15  | 94.08%   | 1247.17     | 177.70        |
|          | 12            | 1587.56       | 1605.31      | 5.48       | 16  | 16  | 81.75%   | 1334.74     | 252.82        |
|          | 9             | 1781.33       | 1781.33      | 5.16       | 16  | 22  | 62.35%   | 1404.64     | 376.69        |
| R104     | 15            | 1196.46       | 1205.25      | 4.49       | 12  | 12  | 97.32%   | 1068.52     | 127.94        |
|          | 12            | 1205.35       | 1207.31      | 4.46       | 12  | 12  | 99.62%   | 1080.40     | 124.95        |
|          | 9             | 1349.61       | 1367.57      | 5.57       | 13  | 13  | 75.18%   | 1134.44     | 215.17        |
| R105     | 15            | 1611.26       | 1620.13      | 5.93       | 17  | 19  | 91.47%   | 1402.09     | 209.17        |
|          | 12            | 1709.35       | 1719.73      | 8.37       | 18  | 18  | 75.24%   | 1427.91     | 281.44        |
|          | 9             | 1844.32       | 1889.96      | 7.17       | 17  | 22  | 60.17%   | 1452.33     | 391.99        |
| R106     | 15            | 1481.15       | 1487.08      | 5.26       | 15  | 16  | 95.94%   | 1292.94     | 188.21        |
|          | 12            | 1585.23       | 1601.81      | 6.05       | 16  | 16  | 83.01%   | 1340.76     | 244.47        |
|          | 9             | 1761.09       | 1768.93      | 8.15       | 17  | 20  | 61.91%   | 1403.60     | 357.49        |
| R107     | 15            | 1281.10       | 1287.66      | 4.80       | 13  | 15  | 95.45%   | 1138.75     | 142.35        |
|          | 12            | 1293.38       | 1303.12      | 4.66       | 12  | 13  | 97.84%   | 1133.48     | 159.90        |
|          | 9             | 1435.18       | 1456.10      | 6.78       | 14  | 14  | 74.43%   | 1204.12     | 231.06        |
| R108     | 15            | 1184.82       | 1192.51      | 4.58       | 12  | 13  | 97.07%   | 1044.58     | 140.24        |
|          | 12            | 1184.82       | 1188.70      | 4.51       | 12  | 13  | 97.07%   | 1044.58     | 140.24        |
|          | 9             | 1326.53       | 1344.89      | 5.92       | 12  | 14  | 81.06%   | 1101.64     | 224.89        |
| R109     | 15            | 1321.65       | 1345.20      | 5.28       | 14  | 14  | 98.34%   | 1187.94     | 133.71        |
|          | 12            | 1439.74       | 1443.44      | 6.31       | 16  | 14  | 81.80%   | 1254.87     | 184.88        |
|          | 9             | 1514.36       | 1549.06      | 8.44       | 15  | 17  | 68.36%   | 1247.77     | 266.59        |
| R110     | 15            | 1223.23       | 1245.82      | 4.73       | 13  | 12  | 97.82%   | 1108.50     | 114.73        |
|          | 12            | 1259.67       | 1282.19      | 4.90       | 12  | 15  | 96.80%   | 1110.69     | 148.98        |
|          | 9             | 1388.32       | 1399.65      | 10.36      | 14  | 13  | 72.59%   | 1169.16     | 219.16        |
| R111     | 15            | 1249.07       | 1260.19      | 4.80       | 13  | 14  | 97.23%   | 1117.56     | 131.51        |
|          | 12            | 1282.89       | 1294.57      | 5.02       | 13  | 15  | 94.49%   | 1133.34     | 149.55        |
|          | 9             | 1400.51       | 1419.57      | 8.63       | 14  | 14  | 73.81%   | 1173.47     | 227.03        |
| R112     | 15            | 1154.92       | 1165.44      | 4.61       | 12  | 14  | 98.61%   | 1035.18     | 119.74        |
|          | 12            | 1154.92       | 1162.49      | 4.55       | 12  | 14  | 98.61%   | 1035.18     | 119.74        |
|          | 9             | 1308.94       | 1322.73      | 7.52       | 12  | 14  | 80.41%   | 1095.59     | 213.35        |
| R201     | 3             | 1107.61       | 1107.98      | 2.73       | 8   | 0   | 73.70%   | 1107.61     | 0.00          |
|          | 2             | 1164.94       | 1168.44      | 4.39       | 7   | 4   | 65.81%   | 1123.62     | 41.32         |
|          | 1             | 1238.11       | 1243.33      | 4.23       | 7   | 7   | 41.25%   | 1142.15     | 95.96         |
| R202     | 3             | 993.99        | 998.95       | 2.46       | 6   | 0   | 70.92%   | 993.99      | 0.00          |
|          | 2             | 1017.17       | 1020.55      | 3.11       | 7   | 1   | 59.14%   | 1013.66     | 3.52          |
|          | 1             | 1068.85       | 1083.50      | 4.10       | 6   | 4   | 36.44%   | 1005.84     | 63.01         |
| R203     | 3             | 875.29        | 880.27       | 2.59       | 6   | 1   | 77.29%   | 874.74      | 0.54          |
|          | 2             | 897.73        | 905.48       | 3.31       | 6   | 3   | 71.47%   | 885.04      | 12.69         |
|          | 1             | 955.61        | 966.21       | 3.40       | 5   | 5   | 52.30%   | 891.58      | 64.03         |
| R204     | 3             | 732.97        | 738.82       | 2.32       | 4   | 0   | 78.15%   | 732.97      | 0.00          |
|          | 2             | 735.91        | 740.79       | 2.31       | 4   | 0   | 74.66%   | 735.91      | 0.00          |
|          | 1             | 772.47        | 776.40       | 2.43       | 6   | 0   | 51.98%   | 772.47      | 0.00          |
| R205     | 3             | 960.41        | 967.76       | 2.66       | 6   | 1   | 80.03%   | 959.70      | 0.71          |
|          | 2             | 968.47        | 977.19       | 3.31       | 6   | 2   | 80.24%   | 967.04      | 1.43          |
|          | 1             | 1043.04       | 1054.77      | 3.69       | 6   | 5   | 51.92%   | 987.80      | 55.24         |
| R206     | 3             | 907.10        | 910.44       | 2.90       | 6   | 2   | 82.35%   | 905.41      | 1.69          |
|          | 2             | 920.09        | 926.70       | 3.03       | 6   | 2   | 76.90%   | 907.90      | 12.19         |
|          | 1             | 1004.85       | 1017.90      | 3.84       | 5   | 4   | 60.22%   | 938.13      | 66.71         |
| R207     | 3             | 795.51        | 811.20       | 2.37       | 5   | 0   | 59.55%   | 795.51      | 0.00          |
|          | 2             | 795.51        | 802.39       | 2.33       | 5   | 0   | 59.55%   | 795.51      | 0.00          |
|          | 1             | 818.25        | 840.36       | 3.04       | 6   | 1   | 50.94%   | 818.08      | 0.17          |
| R208     | 3             | 720.80        | 725.18       | 2.50       | 4   | 1   | 77.59%   | 715.38      | 5.42          |
|          | 2             | 729.22        | 736.50       | 2.36       | 4   | 1   | 83.04%   | 728.60      | 0.62          |
|          | 1             | 765.13        | 768.87       | 3.16       | 4   | 1   | 74.41%   | 750.87      | 14.27         |
| R209     | 3             | 872.20        | 874.94       | 2.30       | 6   | 1   | 79.72%   | 871.86      | 0.34          |
|          | 2             | 873.37        | 879.21       | 3.09       | 6   | 0   | 80.05%   | 873.37      | 0.00          |
|          | 1             | 941.47        | 951.63       | 3.84       | 4   | 5   | 70.28%   | 882.11      | 59.36         |
| R210     | 3             | 842.56        | 849.00       | 2.46       | 5   | 2   | 87.52%   | 839.74      | 2.82          |
|          | 2             | 848.67        | 855.03       | 2.70       | 5   | 2   | 85.65%   | 842.59      | 6.09          |
|          | 1             | 902.14        | 912.62       | 3.45       | 5   | 5   | 52.85%   | 846.18      | 55.96         |
| R211     | 3             | 760.16        | 772.29       | 2.45       | 4   | 0   | 71.52%   | 760.16      | 0.00          |
|          | 2             | 765.78        | 776.09       | 2.40       | 5   | 0   | 57.64%   | 765.78      | 0.00          |
|          | 1             | 793.55        | 804.92       | 2.93       | 5   | 2   | 56.67%   | 789.52      | 4.03          |

NFS: no feasible solution found.

Table B.6: Detailed Results for RC1- and RC2-Type Large-Scale Instances with Different Numbers of Depot Chargers

| Instance | $\mathcal{B}$ | Obj (Best 10) | Obj (Avg 10) | Time (min) | #NV | #CS | Dep. SOC | Travel Time | Charging Time |
|----------|---------------|---------------|--------------|------------|-----|-----|----------|-------------|---------------|
| RC101    | 13            | 1989.67       | 2028.73      | 10.37      | 18  | 18  | 78.54%   | 1751.95     | 237.72        |
|          | 11            | 2099.93       | 2119.02      | 10.34      | 19  | 19  | 66.85%   | 1800.41     | 299.52        |
|          | 9             | 2388.90       | 2388.90      | 6.36       | 20  | 25  | 52.42%   | 1961.16     | 427.74        |
| RC102    | 13            | 1829.51       | 1835.73      | 6.42       | 17  | 15  | 82.21%   | 1632.40     | 197.12        |
|          | 11            | 1894.27       | 1910.96      | 6.42       | 17  | 16  | 71.65%   | 1639.93     | 254.34        |
|          | 9             | 2024.95       | 2048.53      | 9.12       | 18  | 20  | 57.89%   | 1696.00     | 328.95        |
| RC103    | 13            | 1501.30       | 1513.59      | 5.11       | 14  | 13  | 91.96%   | 1370.42     | 130.88        |
|          | 11            | 1559.00       | 1574.01      | 5.82       | 14  | 13  | 84.26%   | 1388.57     | 170.43        |
|          | 9             | 1636.43       | 1647.08      | 11.82      | 15  | 14  | 67.36%   | 1407.53     | 228.90        |
| RC104    | 13            | 1314.85       | 1321.68      | 4.50       | 12  | 13  | 97.39%   | 1209.24     | 105.61        |
|          | 11            | 1368.96       | 1407.03      | 4.65       | 12  | 14  | 94.25%   | 1240.17     | 128.79        |
|          | 9             | 1477.59       | 1487.43      | 8.53       | 12  | 13  | 78.75%   | 1278.08     | 199.51        |
| RC105    | 13            | 1711.76       | 1749.30      | 6.47       | 16  | 17  | 85.65%   | 1541.13     | 170.62        |
|          | 11            | 1804.32       | 1817.08      | 7.35       | 16  | 16  | 73.74%   | 1566.39     | 237.93        |
|          | 9             | 1968.15       | 1993.70      | 11.87      | 17  | 17  | 59.01%   | 1646.34     | 321.81        |
| RC106    | 13            | 1613.91       | 1626.14      | 5.76       | 15  | 15  | 90.22%   | 1466.47     | 147.44        |
|          | 11            | 1674.54       | 1685.81      | 5.46       | 15  | 14  | 79.30%   | 1474.44     | 200.10        |
|          | 9             | 1776.30       | 1801.49      | 7.67       | 15  | 18  | 67.61%   | 1509.72     | 266.58        |
| RC107    | 13            | 1373.73       | 1391.35      | 4.60       | 12  | 13  | 98.57%   | 1255.00     | 118.72        |
|          | 11            | 1474.48       | 1494.33      | 5.46       | 12  | 14  | 93.22%   | 1313.94     | 160.54        |
|          | 9             | 1582.35       | 1599.66      | 11.43      | 14  | 15  | 71.68%   | 1366.85     | 215.50        |
| RC108    | 13            | 1278.60       | 1291.56      | 4.33       | 12  | 13  | 99.58%   | 1188.75     | 89.86         |
|          | 11            | 1337.52       | 1356.37      | 4.42       | 12  | 12  | 95.45%   | 1220.57     | 116.95        |
|          | 9             | 1462.28       | 1466.86      | 7.14       | 13  | 11  | 77.40%   | 1280.42     | 181.85        |
| RC201    | 3             | 1276.53       | 1277.92      | 3.20       | 10  | 2   | 57.29%   | 1268.25     | 8.28          |
|          | 2             | 1316.52       | 1328.06      | 7.87       | 10  | 5   | 49.93%   | 1284.26     | 32.26         |
|          | 1             | 1376.20       | 1410.94      | 7.64       | 9   | 8   | 27.92%   | 1272.31     | 103.89        |
| RC202    | 3             | 1140.10       | 1140.78      | 2.73       | 8   | 0   | 52.18%   | 1140.10     | 0.00          |
|          | 2             | 1149.83       | 1151.18      | 3.45       | 8   | 0   | 52.62%   | 1149.83     | 0.00          |
|          | 1             | 1203.69       | 1221.91      | 5.50       | 8   | 6   | 32.91%   | 1155.67     | 48.03         |
| RC203    | 3             | 962.16        | 968.20       | 2.55       | 6   | 1   | 73.38%   | 957.50      | 4.66          |
|          | 2             | 979.27        | 985.34       | 2.72       | 7   | 2   | 61.55%   | 970.08      | 9.19          |
|          | 1             | 1058.10       | 1066.90      | 3.63       | 6   | 5   | 43.04%   | 994.74      | 63.37         |
| RC204    | 3             | 854.39        | 857.40       | 2.74       | 5   | 3   | 92.96%   | 836.47      | 17.92         |
|          | 2             | 870.66        | 873.66       | 2.90       | 6   | 3   | 75.02%   | 846.40      | 24.26         |
|          | 1             | 952.51        | 961.93       | 3.56       | 5   | 7   | 54.26%   | 869.60      | 82.91         |
| RC205    | 3             | 1079.12       | 1084.05      | 3.01       | 7   | 2   | 78.72%   | 1077.14     | 1.98          |
|          | 2             | 1092.19       | 1094.66      | 3.28       | 7   | 3   | 72.45%   | 1078.44     | 13.75         |
|          | 1             | 1183.08       | 1188.62      | 3.87       | 7   | 7   | 42.49%   | 1104.25     | 78.82         |
| RC206    | 3             | 1073.35       | 1078.32      | 2.46       | 7   | 0   | 66.88%   | 1073.35     | 0.00          |
|          | 2             | 1096.16       | 1104.35      | 2.89       | 7   | 0   | 68.30%   | 1096.16     | 0.00          |
|          | 1             | 1163.05       | 1171.99      | 3.71       | 6   | 6   | 40.32%   | 1093.05     | 70.00         |
| RC207    | 3             | 925.30        | 932.23       | 2.47       | 6   | 0   | 72.66%   | 925.30      | 0.00          |
|          | 2             | 927.06        | 934.33       | 2.81       | 6   | 0   | 72.80%   | 927.06      | 0.00          |
|          | 1             | 1012.31       | 1023.11      | 3.52       | 5   | 6   | 46.41%   | 948.47      | 63.84         |
| RC208    | 3             | 823.63        | 827.87       | 2.47       | 5   | 2   | 89.66%   | 811.05      | 12.58         |
|          | 2             | 823.63        | 830.41       | 2.84       | 5   | 2   | 89.66%   | 811.05      | 12.58         |
|          | 1             | 882.84        | 902.69       | 3.94       | 5   | 4   | 50.11%   | 811.47      | 71.37         |

## Appendix C. Detailed results for the JD Logistics case study

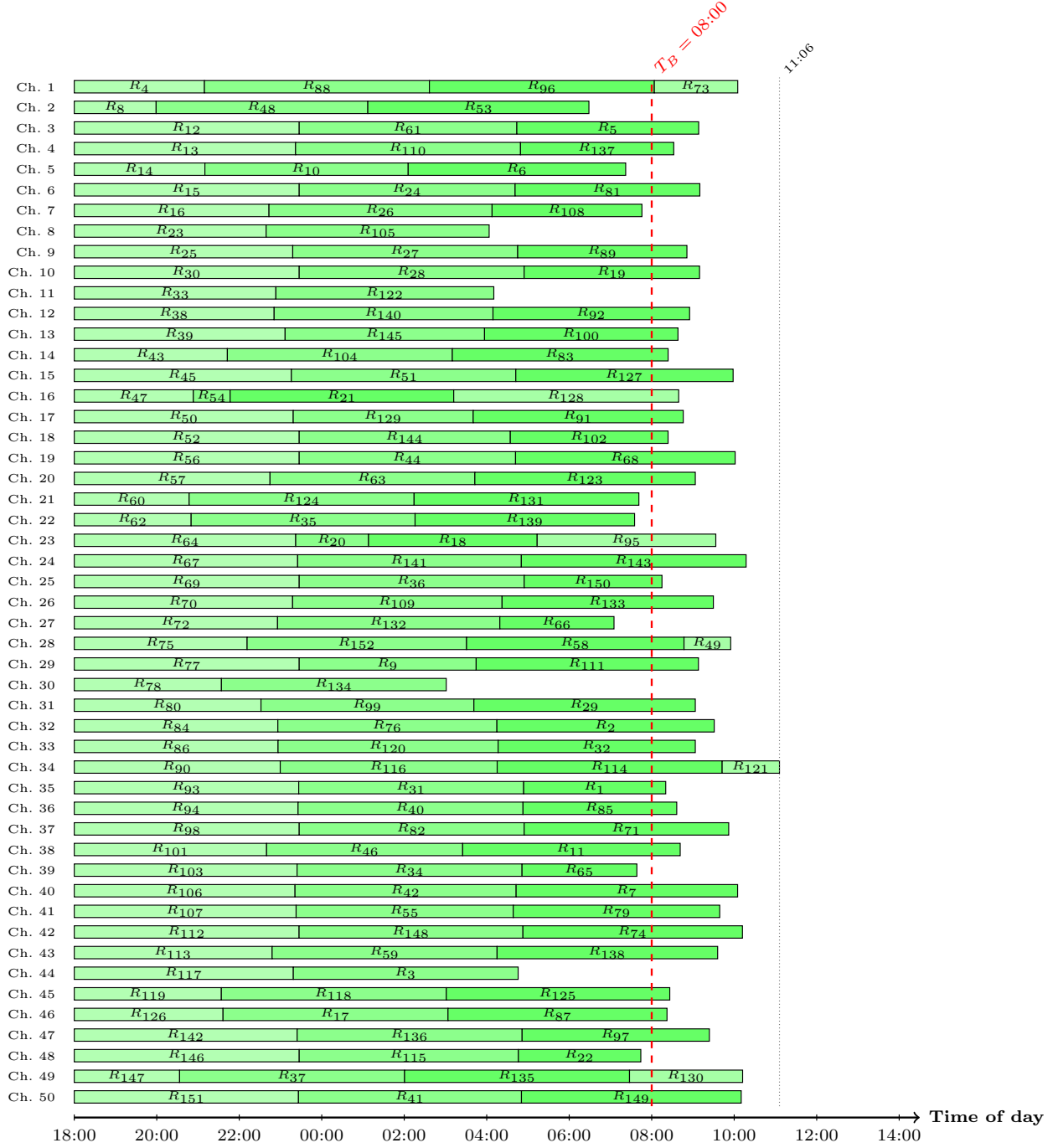


Figure C.1: Depot charging schedule for the JD Logistics case study with  $B = 50$ .

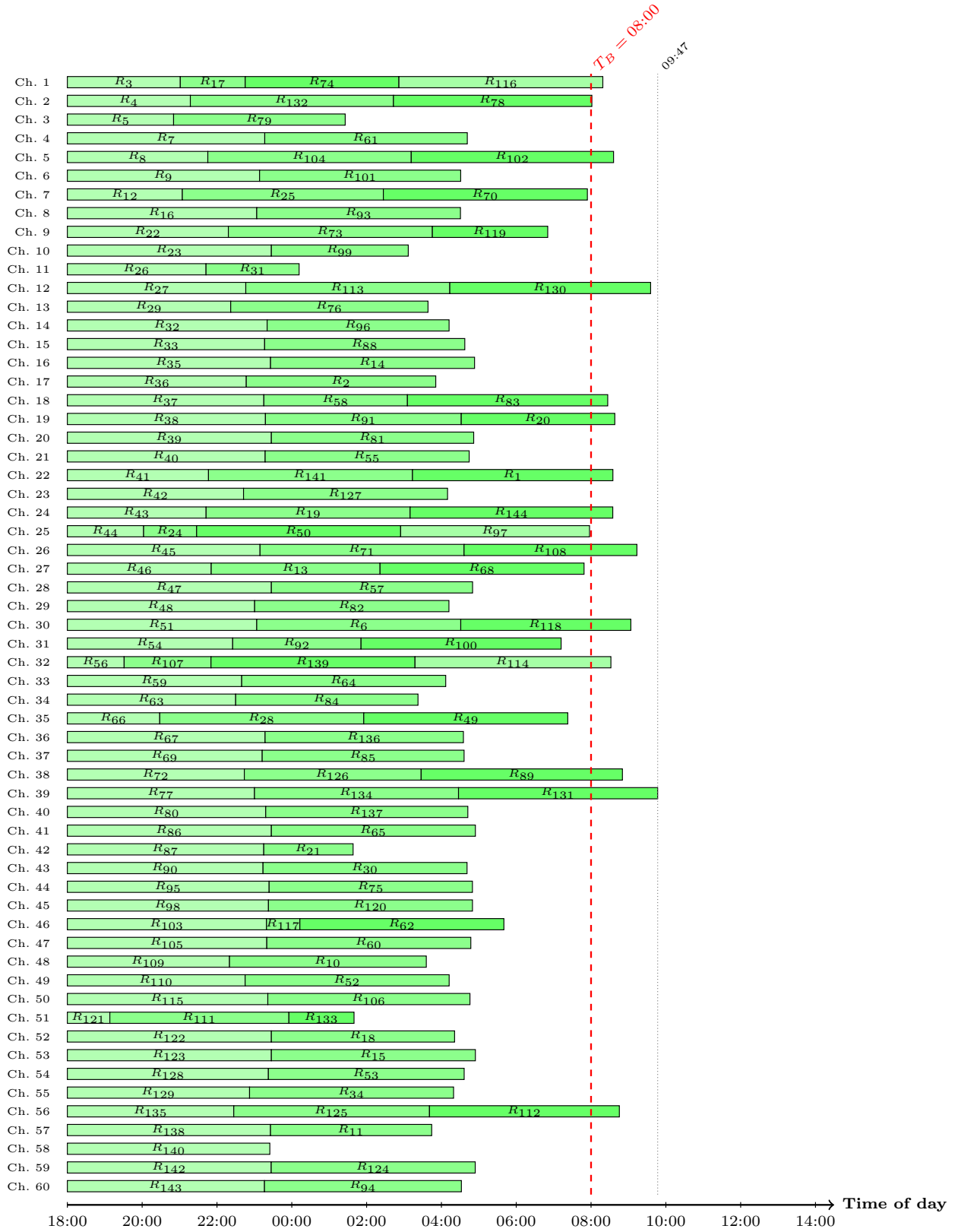


Figure C.2: Depot charging schedule for the JD Logistics case study with  $\mathcal{B} = 60$ .



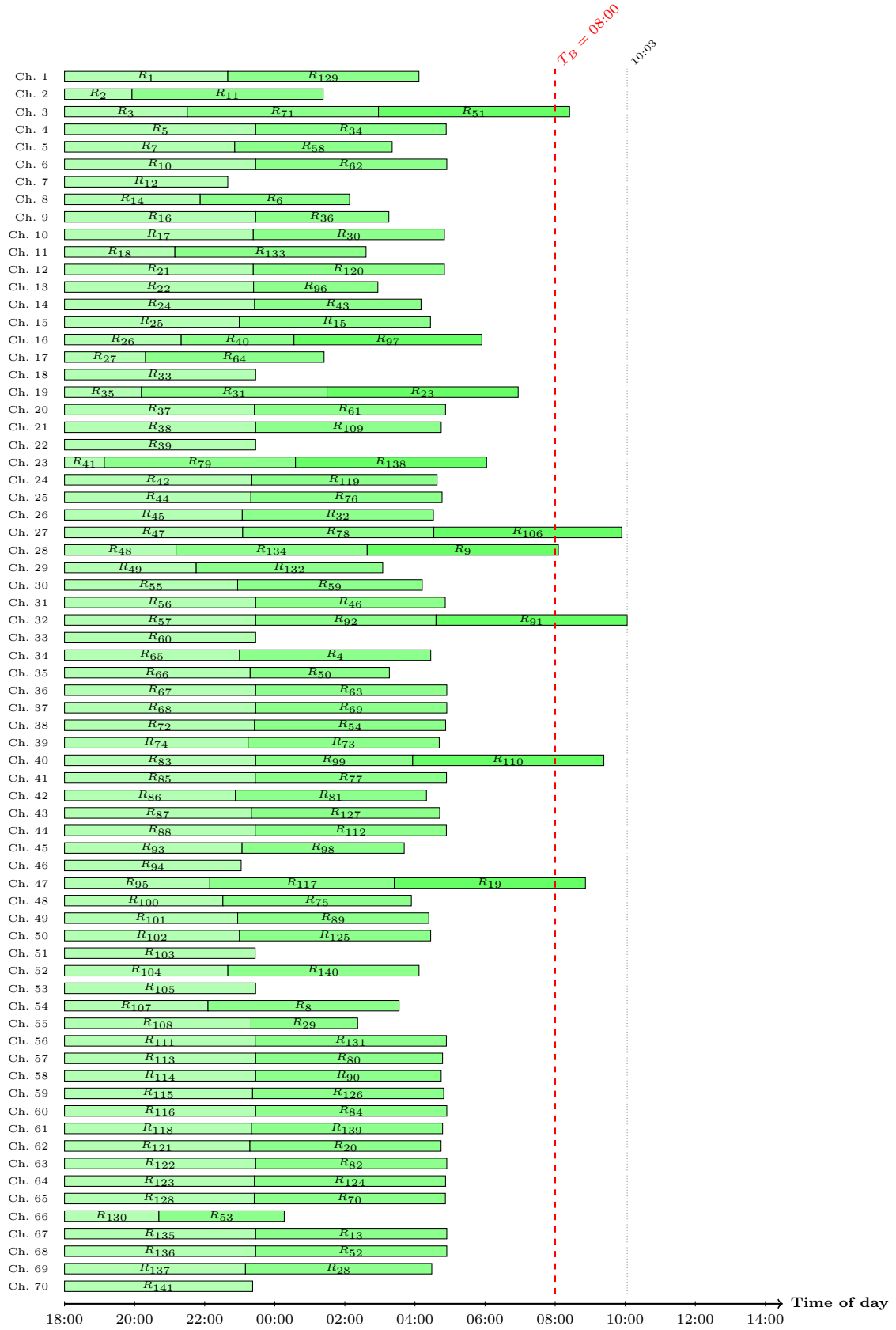


Figure C.3: Depot charging schedule for the JD Logistics case study with  $\mathcal{B} = 70$ .

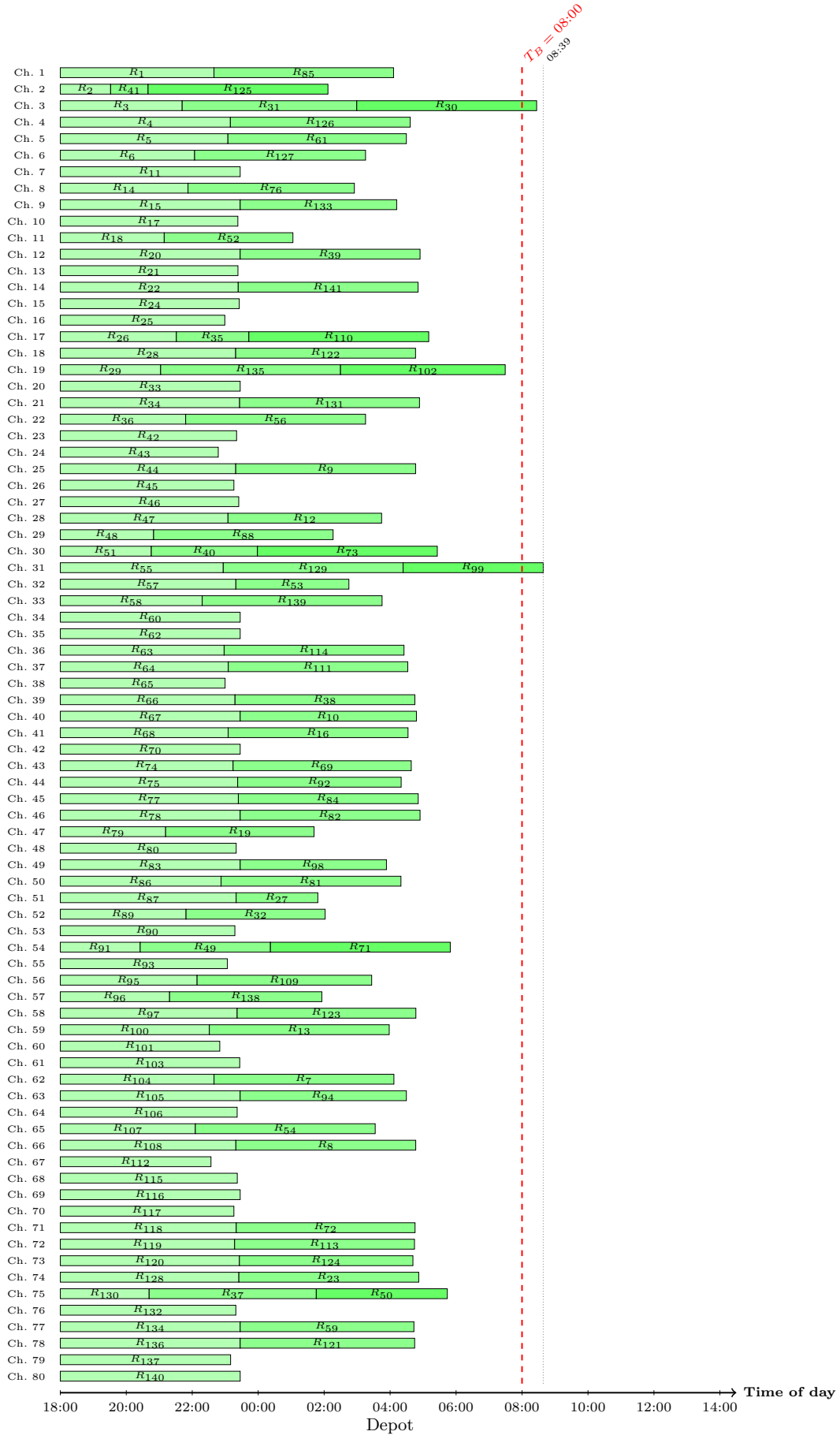


Figure C.4: Depot charging schedule for the JD Logistics case study with  $\mathcal{B} = 80$ .

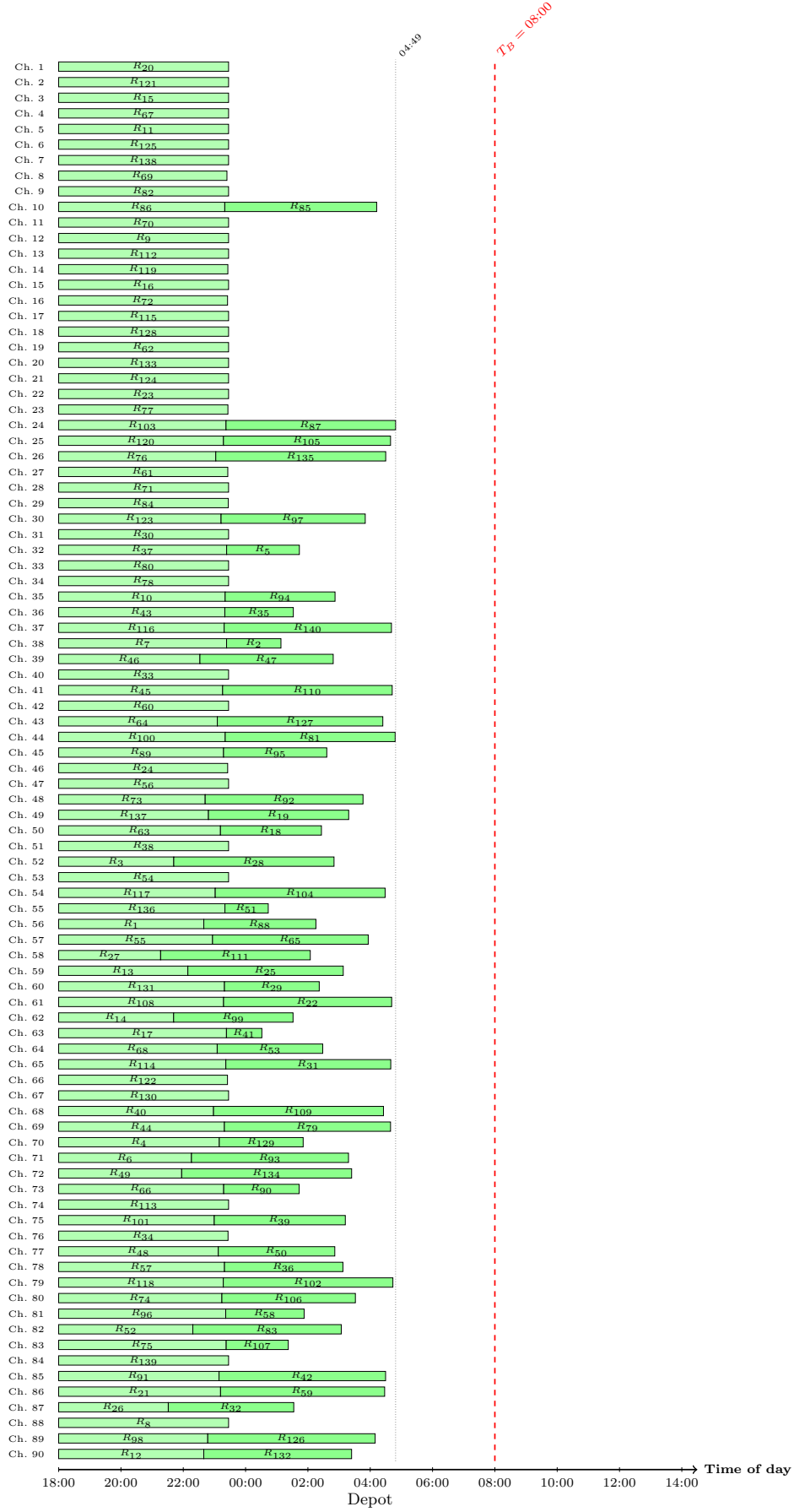


Figure C.5: Depot charging schedule for the JD Logistics case study with  $B = 90$ .

## Appendix D. Adaptation of the matheuristic to the mE-VRSPTW

The matheuristic developed for the EVRP-OC can be adapted to solve the mE-VRSPTW of Yamín et al. (2025) while retaining its general AILS search framework. The main modifications arise from three problem-specific features of the mE-VRSPTW. First, en-route charging is not allowed, so all energy required by a route must be supplied at the depot before departure. Second, the problem is defined on a multigraph, where multiple distance–energy alternatives may exist between two consecutive vertices. Third, depot charging is performed over discrete charging periods according to a piecewise-linear charging function. Accordingly, the public-CS-specific route evaluation and continuous charging decisions of the EVRP-OC are replaced by a multigraph route evaluator and a discrete depot-charging coordinator.

The main customer-sequence search mechanisms of the AILS are retained, including randomized initialization, perturbation, adaptive penalties, and local-search neighborhoods. Since the mE-VRSPTW does not contain public CSs, the ReplacePath operator and the energy-shortage repair procedure used for en-route charging in the EVRP-OC are no longer required. The search therefore operates directly on customer sequences, using standard inter- and intra-route neighborhood moves to modify the customer assignment and visiting order.

For a fixed customer sequence, however, a route does not correspond to a unique path because each pair of consecutive vertices may be connected by multiple multigraph arcs. We therefore generate a set of nondominated realizations, hereafter referred to as *route options*. A forward labeling procedure propagates travel distance, travel time, energy consumption, and time-window feasibility along the customer sequence while considering the available arc alternatives. Dominated labels and route options are discarded. Each retained option  $o$  of route  $r$  is characterized by its total routing distance  $d_{ro}$ , energy requirement  $e_{ro}$ , latest feasible departure time  $T_{ro}^{\max}$ , and the corresponding overnight charging duration.

Because all energy must be supplied before departure, the charging duration of a route option is obtained from the inverse piecewise-linear charging function of the mE-VRSPTW. Following its discrete-time charging representation, the resulting duration is converted into the required number of complete charging periods. Hence, selecting a route option simultaneously determines its routing cost and the amount of scarce depot-charging capacity required before departure.

During the AILS, a route option is evaluated using a search criterion that accounts for both routing distance and overnight charging demand. In particular, the charging-duration component is weighted by an adaptive parameter  $\lambda$ . When the current route set is difficult to accommodate within the available depot charging capacity,  $\lambda$  is increased, placing greater emphasis on route options with lower energy requirements and shorter charging durations. When the depot charging schedule is readily feasible,  $\lambda$  is reduced, allowing the search to place greater emphasis on routing distance. The original mE-VRSPTW objective remains the minimization of total routing distance; the charging-related term is used only to guide the heuristic search. This mechanism plays a role analogous to the adaptive departure-SOC control used for the EVRP-OC.

For a selected route option, the corresponding overnight charging operation is represented by its charging duration and latest feasible completion time. The depot subproblem is therefore a nonpreemptive scheduling problem on  $B$  identical chargers, where each EV must be assigned to exactly one charger and its charging operation must be completed before the departure deadline of the selected route option. To evaluate candidate solutions efficiently during the AILS, we first construct a greedy earliest-deadline schedule. If this procedure successfully schedules all charging operations, the resulting schedule provides an explicit certificate of depot-charging feasibility. Otherwise, a bounded depth-first search is used to explore alternative charger assignments. Reaching the prescribed search limit is treated as an unresolved case rather than as proof of infeasibility; a solution is classified as depot-feasible only when a complete feasible charging schedule has been explicitly constructed.

The route-pooling idea of the original matheuristic is also retained, but the continuous SP model developed for the EVRP-OC is not transferred literally. Instead, the coordination step jointly selects multigraph route options for the customer routes and verifies their compatibility with the discrete depot-charging schedule. For small combinations of route options, the alternatives can be examined directly. For larger combinations, the implementation explores promising option selections within prescribed computational limits while retaining only solutions for which a feasible depot schedule is explicitly constructed.

Finally, when a depot-feasible solution is about to improve the global incumbent, an additional route-option refinement is performed. Starting from the validated solution, alternative combinations of nondominated route options are explored in increasing order of routing distance. A neighboring state is obtained by replacing the option of one route with another nondominated alternative, allowing coordinated changes across several routes. The incumbent objective provides an upper bound for pruning option combinations that cannot improve the current best solution. The refinement is strictly bounded by the number of explored option states, charger-assignment search effort, and computing time. If a limit is reached, the previously validated incumbent is retained.

Overall, the adaptation preserves the route-generation and intensification philosophy of the EVRP-OC matheuristic while replacing the continuous departure-SOC and public-charging decisions by multigraph route-option selection and discrete overnight charging

coordination. This enables the same general framework to exploit the distance–energy trade-off specific to the mE-VRSPTW while explicitly accounting for the limited charging capacity at the depot.

## References

Yamín, D., Desaulniers, G., & Mendoza, J. E. (2025). The electric vehicle routing and overnight charging scheduling problem on a multigraph. *INFORMS Journal on Computing*, 37, 808–830.